Course: ECE129
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Surflex: Quantifying Surfboard Flex

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The Problem

Surfboards are characterized by geometric measurements which do not fully describe a surfboard's dynamic behavior while surfing.

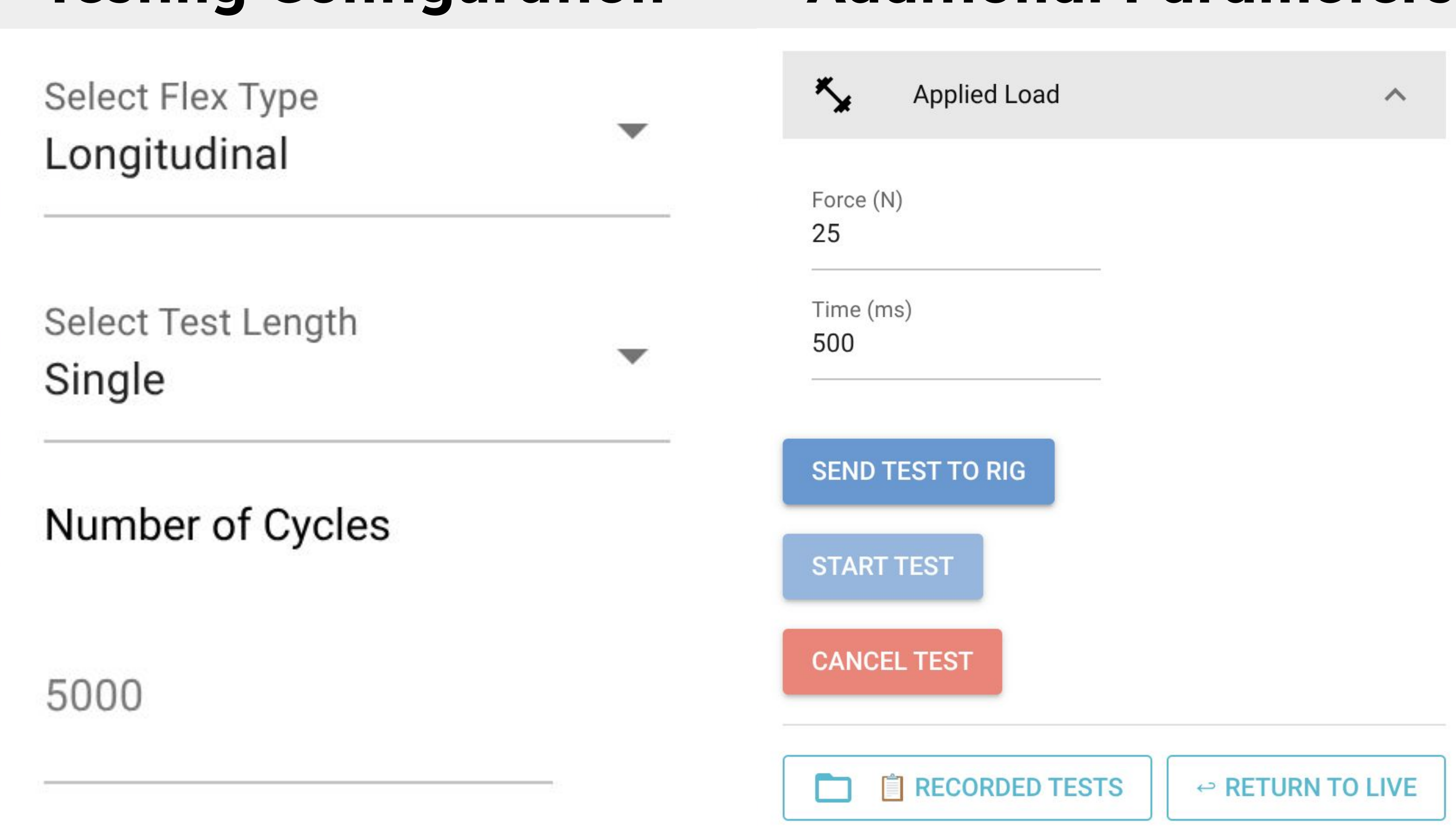
Flex affects a surfboard's feel and performance, but shapers lack a way to reliably communicate and quantify flex behavior to consumers.

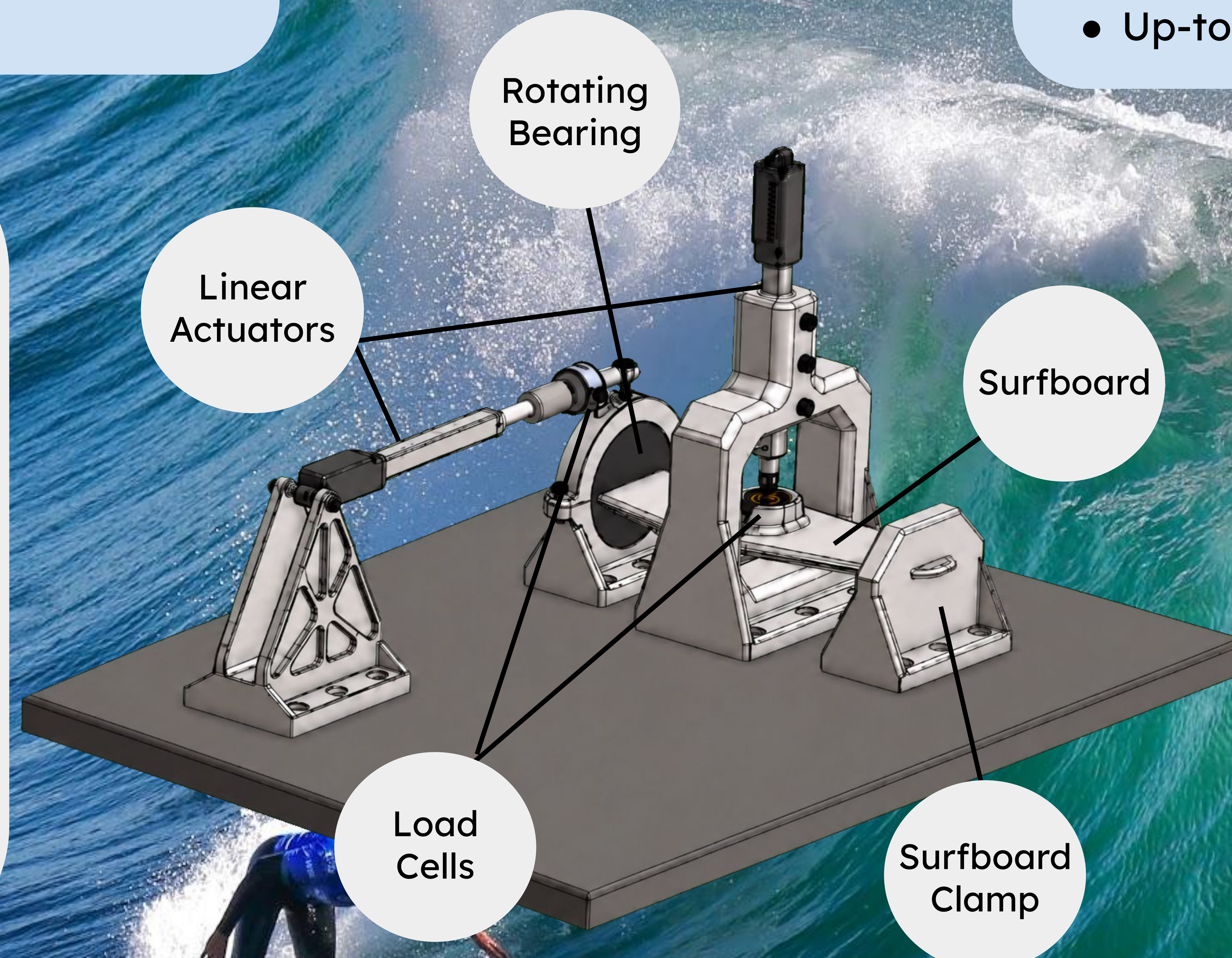
Our Solution

Surflex is a 1:10 scale actuator-based testing rig that allows user-control over test protocols and provides:

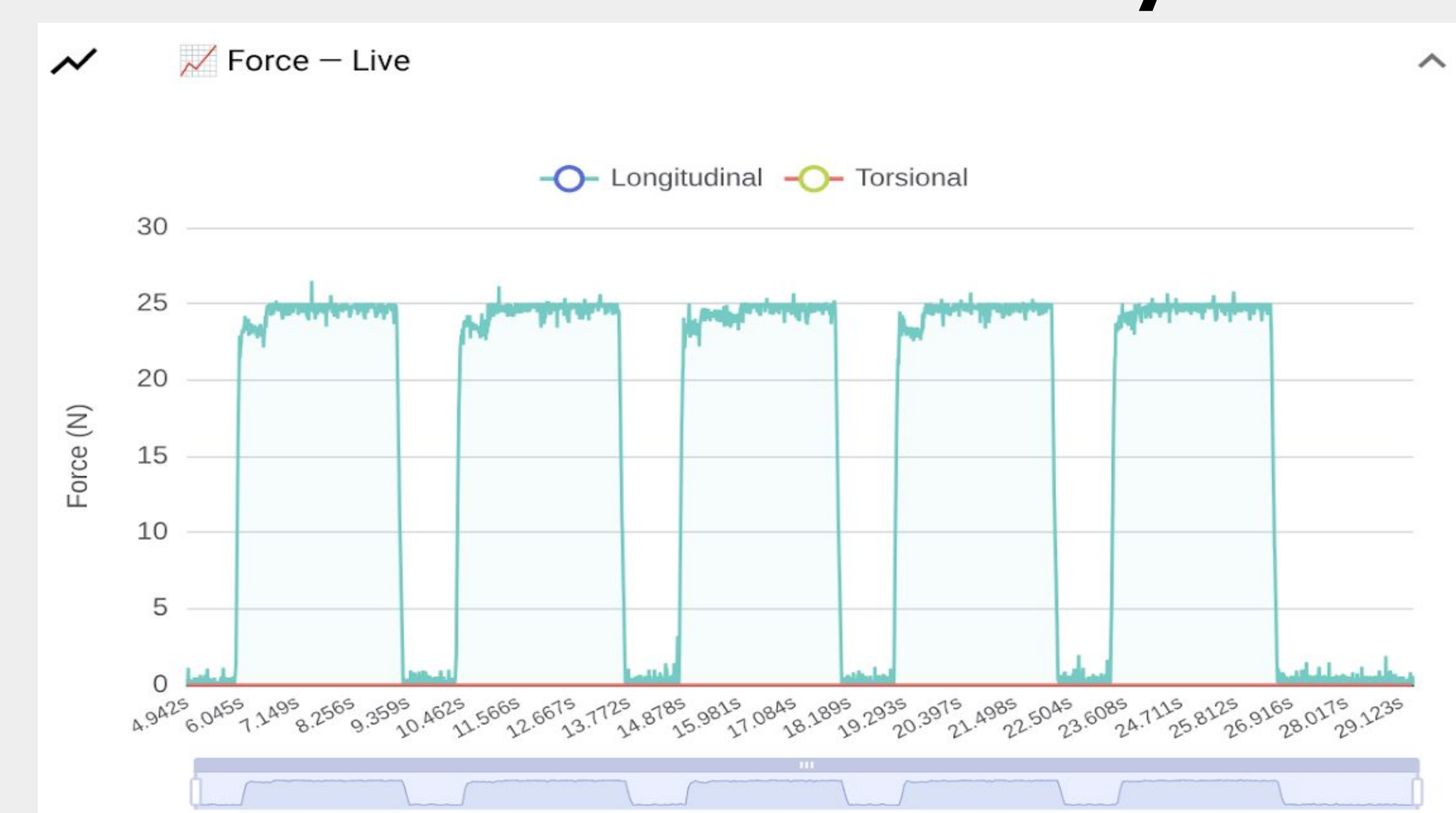
- Real-time data of a surfboard's flex behavior.
- Calculated surfboard stiffness and change over a board's lifetime.
- Up-to-date metrics for consumers and shapers.

GUI Control Center



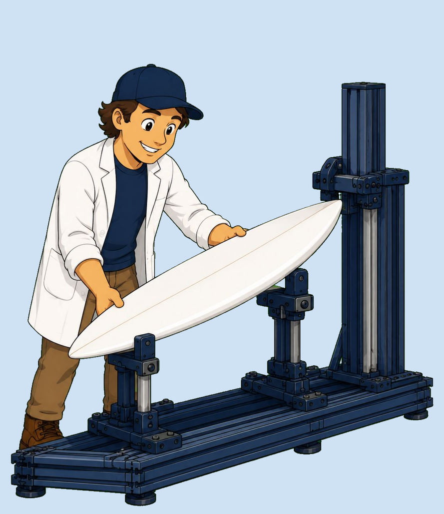


GUI Test Summary



Real-Time Data Scrollbar

How It Works



An operator mounts a scaled surfboard

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Test parameters selected and starts test

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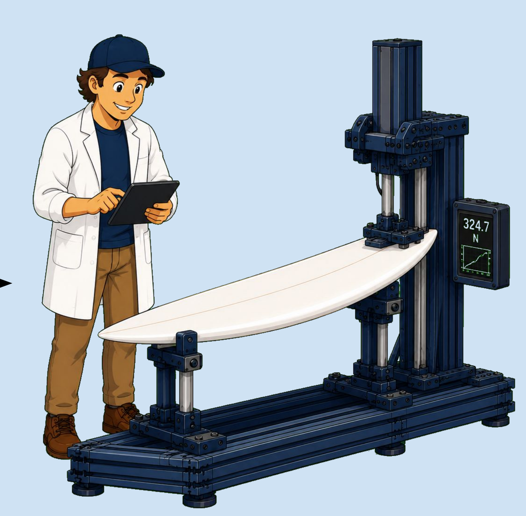

Surfboard bends, measuring deflection and applied force

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Live data is graphed, flex metrics displayed on UI

Output Data Metrics

- Spring Constant
- Torque Constant
- Stiffness Rate
- Rotational Stiffness Rate

Supported Test Modes

- Single/Multi-cycle longitudinal bending
- Single/Multi-cycle torsional twisting
- Longitudinal failure
- Torsional failure

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Future Implementation

1. Develop a full-scale Surflex rig to load surfboards between 5 ft and 11 ft in length.
2. Develop a machine learning model trained on Surflex data.
3. Use the ML model to predict 3D-printed surfboard flex from material properties and infill patterns.